AI CONCEPTS & PRACTICE SERIES

Understanding RAG: Study Guide & Sheet

1. Core Summary: What is RAG?

Retrieval-Augmented Generation (RAG) is a framework designed to improve AI responses by referencing external, trusted data sources before generating an answer. Instead of forcing the Artificial Intelligence model to rely strictly on static memory, RAG acts like an *open-book exam*: it searches a knowledge base for verified information and incorporates those facts into the prompt.

2. Key Terminology Breakdown

R – Retrieval

Searching a database to find relevant documents or text chunks matching the user's specific request.

A – Augmented

Injecting the retrieved text into the AI model's instructions (prompt) to provide immediate context.

G – Generation

Producing the final clear answer using strictly the factual context retrieved from the database.

Embedding Vector

A list of numbers (math representation) that captures the underlying *meaning* of a piece of text.

3. Matching Meanings with Cosine Similarity

Computers compare text meaning using **Cosine Similarity**, a mathematical metric that calculates the angle between two embedding vectors in numerical space:

- **Score near 1.0 (High Match):** Vectors point in almost the same direction. For instance, *"How to transfer cash on app"* and *"Mobile app daily transfer limit"* share high similarity.
- **Score near 0.0 (No Match):** Vectors are unrelated, such as *"How to cook Egusi soup"* versus *"Central Bank of Nigeria transfer limits"*.

4. Real-World Nigerian Scenario

Customer Prompt: "What is the daily bank transfer limit on the app?"

RAG Process: The system converts the prompt into an embedding vector, uses Cosine Similarity to find the official CBN Circular in the bank's vector database (Match Score: 0.93), augments the prompt with the official rule, and generates an exact response: *"According to CBN guidelines, Tier 3 daily mobile limits are capped at ₦5,000,000."*

INTERACTIVE ASSESSMENT

RAG Practice Quiz & Exercise

Practice Questions

Test your understanding of RAG principles, vectors, and cost benefits using these questions.

Question 1: What does the "A" in RAG stand for, and what does it do?

A. Automated: Automatically trains the AI on new internet pages.

B. Augmented: Adds retrieved facts into the prompt before sending it to the AI.

C. Algorithm: Converts user text into numbers using computer algorithms.

Question 2: Why are words converted into Embedding Vectors in RAG?

A. To compress document file sizes for storage.

B. To allow computers to search for text based on conceptual meaning rather than exact spelling.

C. To translate English sentences into other languages automatically.

Question 3: A query vector and document vector have a Cosine Similarity score of 0.92. What does this mean?

A. The document is highly relevant and shares almost the same meaning as the query.

B. The document is completely unrelated to the query.

C. The system encountered an error while calculating the math.

Question 4: How does RAG help reduce LLM API costs for businesses?

A. It completely removes the need to use AI models.

B. It sends only small, relevant document chunks in the prompt instead of huge 50,000-token files.

C. It reduces internet data charges when connecting to server databases.

Answer Key & Explanations

1. **1 – B (Augmented):** Augmentation attaches retrieved documents directly to the user prompt context.
2. **2 – B (Conceptual Meaning):** Vectors capture semantic meaning so "Jollof Rice" matches "Food Order".
3. **3 – A (High Relevance):** Scores close to 1.0 indicate vectors pointing in nearly identical directions.
4. **4 – B (Lower API Tokens):** Sending precise 500-token chunks drastically cuts down input token billing.